

Лабораторная работа

Номер 1

Кобзев Д. К.

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Российский университет дружбы народов, Москва, Россия

Информация

- Кобзев Дмитрий Константинович
- Студент
- Российский университет дружбы народов
- НПИбд-01-23

Целью данной работы является развёртывание в системе виртуализации mininet, знакомство с основными командами для работы с Mininet через командную строку и через графический интерфейс.

Настройка образа VirtualBox

Запускаем систему виртуализации и импортируем файл .ovf. Переходим в настройки системы виртуализации и уточняем параметры настройки виртуальной машины. Запускаем виртуальную машину с Mininet (Рис. 1.1).

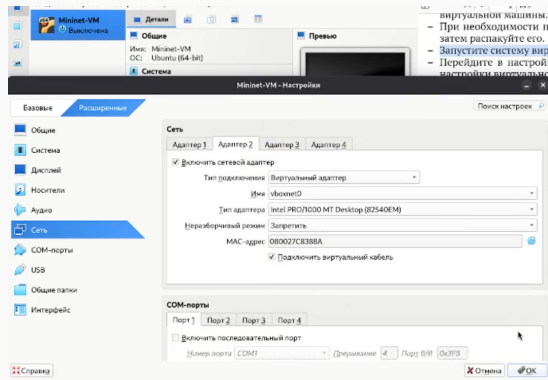


Рис. 1: Настройка образа VirtualBox

Подключение к виртуальной машине

Логинемся в виртуальной машине. Смотрим адрес машины (Рис. 1.2).

```
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.56.101  netmask 255.255.255.0  broadcast 192.168.56.255
    ether 00:00:27:c8:38:8a  txqueuelen 1000  (Ethernet)
    RX packets 2  bytes 1180 (1.1 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 2  bytes 684 (684.0 B)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 128  bytes 9816 (9.8 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 128  bytes 9816 (9.8 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

Рис. 2: Адрес машины

Настроим ssh-подсоединение по ключу к виртуальной машине. Подключаемся к виртуальной машине (Рис. 1.3).

```
mininet@mininet-vm:~$ logout
Connection to 192.168.56.101 closed.
dkkobzev@realme:~$ cd /home/dkkobzev/.ssh
dkkobzev@realme:~/.ssh$ ssh-copy-id mininet@192.168.56.101
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed:
ssh-add -L
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new
key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed --
if you are prompted now it is to install the new keys
mininet@192.168.56.101's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'mininet@192.168.
56.101'"
and check to make sure that only the key(s) you wanted were ad
ded.

dkkobzev@realme:~/.ssh$ ssh -Y mininet@192.168.56.101
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_
64)

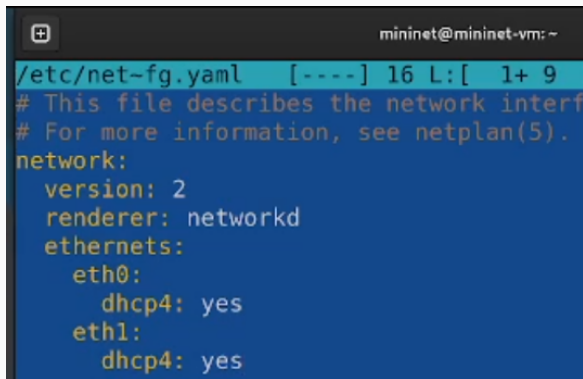
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Failed to connect to https://changelogs.ubuntu.com/meta-releas
e-lts. Check your Internet connection or proxy settings

Last login: Sat Sep 12 01:02:20 2026 from 192.168.56.1
```

Рис. 3: Подключение к виртуальной машине

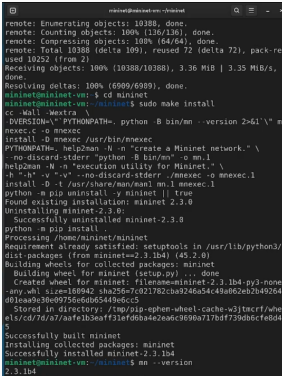
Добавляем для mininet указание на использование двух адаптеров при запуске (Рис. 1.4).



```
mininet@mininet-vm: ~  
/etc/netcfg.yaml [----] 16 L:[ 1+ 9  
# This file describes the network interf  
# For more information, see netplan(5).  
network:  
  version: 2  
  renderer: networkd  
  ethernets:  
    eth0:  
      dhcp4: yes  
    eth1:  
      dhcp4: yes
```

Рис. 4: Настройка доступа к Интернет

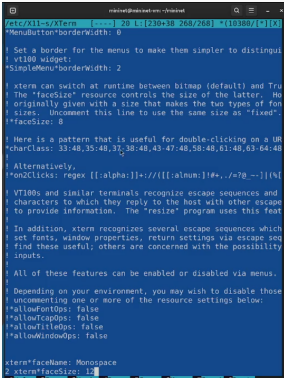
Обновляем версию Mininet (Рис. 1.5).



```
mininet@mininet-vm: ~/mininet
remote: Enumerating objects: 10388, done.
remote: Counting objects: 100% (136/136), done.
remote: Compressing objects: 100% (64/64), done.
remote: Total 10388 (delta 109), reused 72 (delta 72), pack-re
used 10252 (from 2)
Receiving objects: 100% (10388/10388), 3.36 MiB | 3.35 MiB/s,
done.
Resolving deltas: 100% (6909/6909), done.
mininet@mininet-vm: $ cd mininet
mininet@mininet-vm: ~/mininet$ sudo make install
cc -Wall -Wextra \
-DVERSION="\" PYTHONPATH=. python -B bin/mn --version 2>&1\" m
nexec.c -o mnexec
install -o mnexec /usr/bin/mnexec
PYTHONPATH=. help2man -N -n "create a Mininet network." \
--no-discard-stderr "python -B bin/mn" -o mn.1
help2man -N -n "execution utility for Mininet." \
-h "h" -v "-v" --no-discard-stderr ./mnexec -o mnexec.1
install -D -t /usr/share/man/man1 mn.1 mnexec.1
python -m pip uninstall -y mininet || true
Found existing installation: mininet 2.3.0
Uninstalling mininet-2.3.0:
  Successfully uninstalled mininet-2.3.0
python -m pip install .
Processing /home/mininet/mininet
Requirement already satisfied: setuptools in /usr/lib/python3/
dist-packages (from mininet==2.3.1b4) (45.2.0)
Building wheels for collected packages: mininet
  Building wheel for mininet (setup.py) ... done
  Created wheel for mininet: filename=mininet-2.3.1b4-py3-none
-any.whl size=16942 sha256=7c021782cba9246a54c49a062eb2b49264
d01ea9e30e097566db65449e6cc5
  Stored in directory: /tmp/pip-ephem-wheel-cache-w3jtcrf/whe
els/cd/7d/a7/aaf61b3eaff31efdb6ba4e2ea6c9699a717bd7739db6cfe8d4
5
Successfully built mininet
Installing collected packages: mininet
Successfully installed mininet-2.3.1b4
mininet@mininet-vm: ~/mininet$ mn --version
2.3.1b4
```

Рис. 5: Обновление версии Mininet

Увеличиваем размер шрифта и применяем векторные шрифты вместо растровых (Рис. 1.6).



```
mininet@mininet-vm: ~/xterm
/etc/X11-xterm/xterm [----] 20 L:[230+30 268/268] *(10380/1*)[X]
*MenuBar*borderWidth: 0

! Set a border for the menus to make them simpler to distinguish
! vt100 widget:
*SimpleMenu*borderWidth: 2

! xterm can switch at runtime between bitmap (default) and TrueType
! The "faceSize" resource controls the size of the latter.  Here
! originally given with a size that makes the two types of font
! sizes.  Uncomment this line to use the same size as "fixed".
!*faceSize: 8

! Here is a pattern that is useful for double-clicking on a URL
*charClass: 33:48,35:48,37:38:48,43:47:48,58:48,61:48,63:64:48
!
! Alternatively,
!*on2Clicks: regex [[:alpha:]]+://([[:alnum:]]#+./=?@_~)|[%]

! VT100s and similar terminals recognize escape sequences and
! characters to which they reply to the host with other escape
! to provide information.  The "resize" program uses this feature.
!
! In addition, xterm recognizes several escape sequences which
! set fonts, window properties, return settings via escape sequences
! find these useful; others are concerned with the possibility of
! inputs.
!
! All of these features can be enabled or disabled via menus.
!
! Depending on your environment, you may wish to disable those
! uncommenting one or more of the resource settings below:
!*allowFontOps: false
!*allowTcapOps: false
!*allowTitleOps: false
!*allowWindowOps: false

xterm*faceName: Monospace
2 xterm*faceSize: 12
```

Рис. 6: Настройка параметров XTerm

Настройка соединения X11 для суперпользователя

Настраиваем соединения X11 для суперпользователя (Рис. 1.7).

```
mininet@mininet-vm:~/mininet$ xauth list $DISPLAY
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  4632faa77ab529c0b80cd3
749c4af075
mininet@mininet-vm:~/mininet$ sudo -i
root@mininet-vm:~# xauth list
xauth:  file /root/.Xauthority does not exist
root@mininet-vm:~# xauth add mininet-vm/unix:10  MIT-MAGIC-COO
KIE-1  4632faa77ab529c0b80cd3749c4af075
xauth:  file /root/.Xauthority does not exist
root@mininet-vm:~# xauth add mininet-vm/unix:10 MIT-MAGIC-COOK
IE-1 4632faa77ab529c0b80cd3749c4af075
root@mininet-vm:~# xauth list $DISPLAY
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  4632faa77ab529c0b80cd3
749c4af075
```

Рис. 7: Настройка соединения X11 для суперпользователя

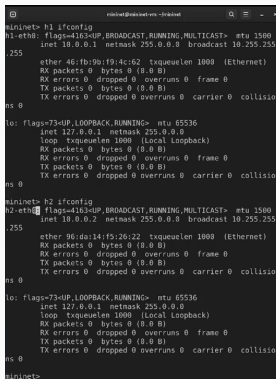
Работа с Mininet с помощью командной строки

Запускаем Mininet с минимальной топологией. Отображаем доступные узлы. Отображаем связи между устройствами в Mininet (Рис. 1.8).

```
mininet> nodes
available nodes are:
c0 h1 h2 s1
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
```

Рис. 8: Работа с Mininet с помощью командной строки

Выполняем команду `ifconfig` на хосте `h1` и `h2` и смотрим интерфейсы хостов (Рис. 1.9).



```
mininet@mininet-vms -br/>mininet> h1 ifconfig
h1-eth8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 46:fb:9b:f0:4c:62 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collision 0
    ns 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collision 0
    ns 0

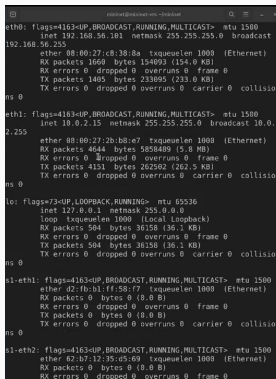
mininet> h2 ifconfig
h2-eth8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 96:da:14:f5:26:22 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collision 0
    ns 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collision 0
    ns 0

mininet>
```

Рис. 9: Интерфейсы хостов `h1` и `h2`

Смотрим интерфейсы коммутатора s1 (Рис. 1.10).



```
mininet@mininet-vty:~$brctlshow
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.101 netmask 255.255.255.0 broadcast
    192.168.56.255
    ether 08:00:27:c8:38:8a txqueuelen 1000 (Ethernet)
    RX packets 1660 bytes 154093 (154.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1405 bytes 233095 (233.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisio
ns 0
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.
    2.255
    ether 08:00:27:2b:b8:e7 txqueuelen 1000 (Ethernet)
    RX packets 4644 bytes 5858489 (5.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4151 bytes 262902 (262.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisio
ns 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 504 bytes 36158 (36.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 504 bytes 36158 (36.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisio
ns 0
s1-eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether d2:fb:b1:ff:58:f7 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisio
ns 0
s1-eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether f2:17:12:38:a5:69 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
```

Рис. 10: Интерфейсы коммутатора

Проверяем соединение между хостами h1 и h2 (Рис. 1.11).

```
mininet> h1 ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.20 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.306 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.048 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.050 ms
^C
--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3029ms
rtt min/avg/max/mdev = 0.048/0.650/2.199/0.900 ms
```

Рис. 11: Проверка связности

Построение и эмуляция сети в Mininet с использованием графического интерфейса

В терминале виртуальной машины mininet запускаем MiniEdit. Добавляем два хоста и один коммутатор, соединяем хосты с коммутатором. Настраиваем IP-адреса на хостах h1 и h2 (Рис. 1.12).

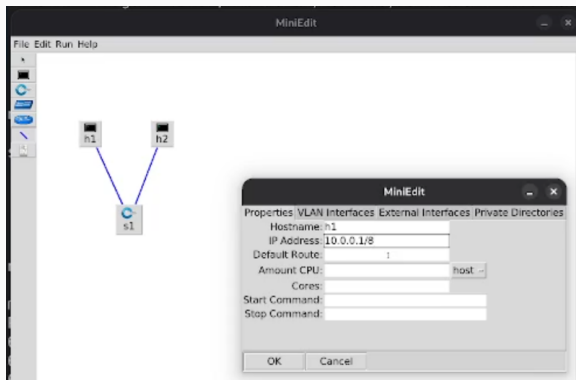


Рис. 12: Построение топологии сети

Построение и эмуляция сети в Mininet с использованием графического интерфейса

Запускаем эмуляцию. Смотрим назначенные хостам адреса (Рис. 1.13).



```
"Host: h1"
root@mininet-vx:/home/mininet/mininet# ifconfig
h1-eth0: flags=4096<UP,BROADCAST,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 52:3d:1c:b4:30:56 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1139 bytes 352920 (352.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1139 bytes 352920 (352.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vx:/home/mininet/mininet# ]

"Host: h2"
root@mininet-vx:/home/mininet/mininet# ifconfig
h2-eth0: flags=4096<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether c2:1a:63:06:04:1b txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1122 bytes 352460 (352.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1122 bytes 352460 (352.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vx:/home/mininet/mininet# ]
```

Рис. 13: Назначенные хостам адреса

Построение и эмуляция сети в Mininet с использованием графического интерфейса

Смотрим интерфейсы коммутатора (Рис. 1.14).

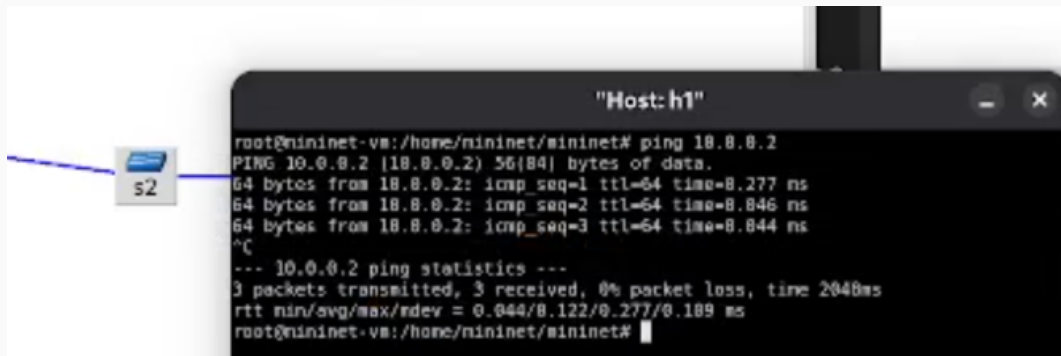


Рис. 14: Проверка связности

Настраиваем автоматическое назначение IP-адресов (Рис. 1.15).

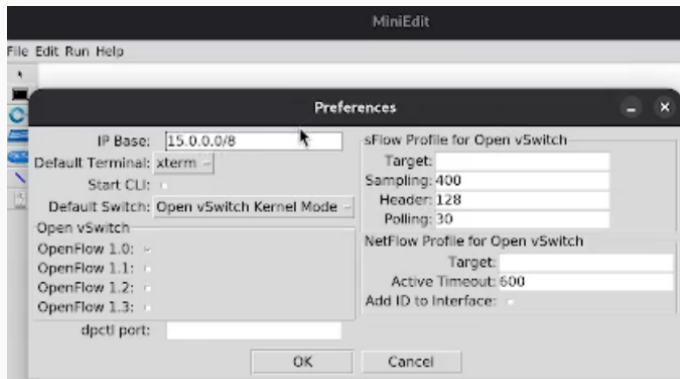
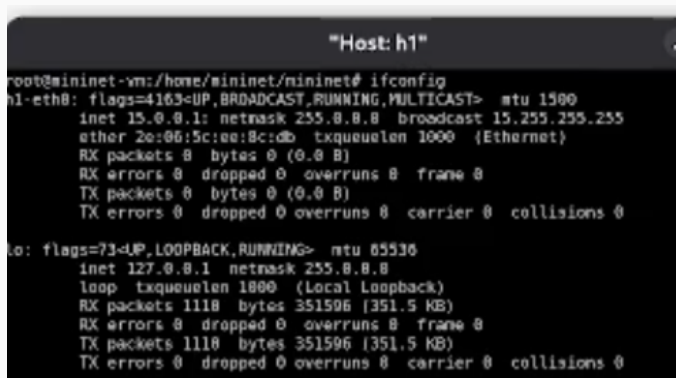


Рис. 15: Автоматическое назначение IP-адресов

Отображаем IP-адреса, назначенные хосту h1 (Рис. 1.16).



```
Host: h1
root@mininet-vm:/home/mininet/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 15.0.0.1 netmask 255.0.0.0 broadcast 15.255.255.255
    ether 2e:06:5c:ee:8c:db txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1110 bytes 351596 (351.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1110 bytes 351596 (351.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рис. 16: IP-адреса, назначенные хосту h1

Сохраняем топологию Mininet (Рис. 1.17).

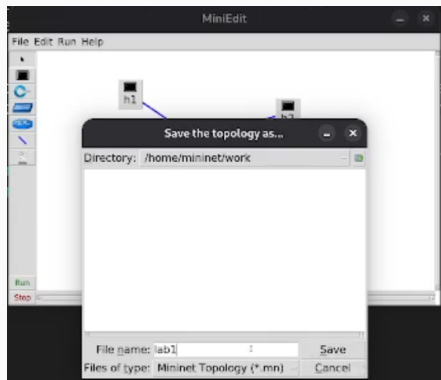


Рис. 17: Сохранение топологии Mininet

В результате выполнения лабораторной работы мною был развернут в системе виртуализации mininet, знакомлен с основными командами для работы с Mininet через командную строку и через графический интерфейс.